Detection of Phishing Websites Using Machine Learning

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Introduction and Problem Statement

- Phishing is a type of cyber attack that uses social engineering and technical fraud to steal sensitive data from consumers, such as personal information and financial account credentials[1].
- In recent years, researchers have attempted to detect phishing websites employing a variety of approaches based on machine learning classifiers. Those classifiers can be listed as URL-based[2], content-based[3], and hybrid-based[4].

1. Anti-Phishing Working Group. Phishing activity trends report. pages 1–11, June 2020
2. O. K. Sahingoz, E. Buber, O. Demir, and B. Diri. Machine learning based phishing detection from urls. Expert Systems with Applications, 117:345–357, 2019.
3. H. Zhang, G. Liu, T. W. Chow, and W. Liu. Textual and visual content-based anti-phishing: a bayesian approach. IEEE Transactions on Neural Networks, 22(10):1532–1546, 2011.
4. W. Zhang, Q. Jiang, L. Chen, and C. Li. Two-stage elm for phishing web pages detection using hybrid features. World Wide Web, 20(4):797–813, 2017.

Solution Approach

Dataset Collection

- Sources: OpenPhish and Alexa Top 1000 websites
  - 698 urls in total (0.8 training ratio)
- Storage in MongoDB
- Google Cloud Virtual Machine (VM)
  - Firewall configuration
  - Only outbound HTTP/HTTPS traffic was permitted
  - No sensitive data was stored
- Selenium Web scraping

Feature Extraction

- URL-based features
  - Length
  - Entropy
  - hyphen count
  - squatting (and variations)
- HTML-based features
  - `<input>` tags
  - `<href>` tags
  - `<embed> tags`
  - `Internal/external links`
  - Suspicious words
- Python libraries like `pandas`, `sklearn`, and `shap`.

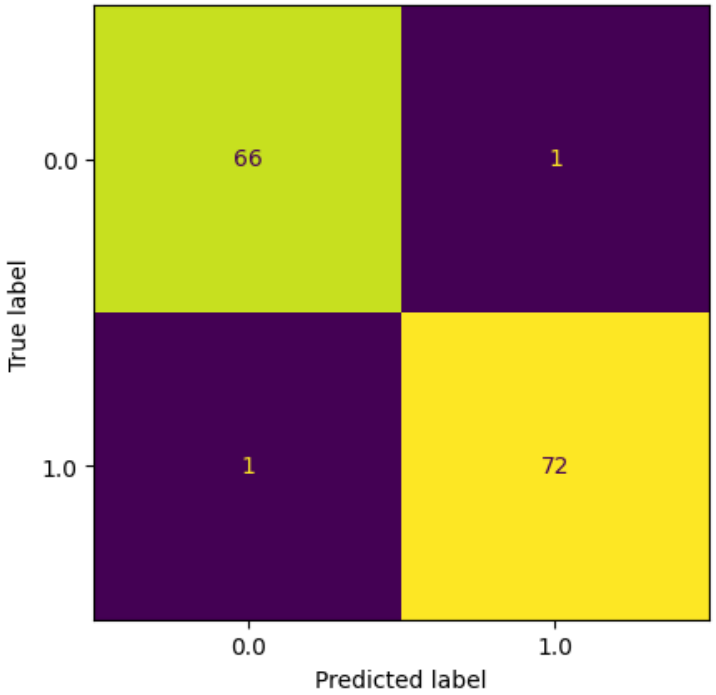
Models and Evaluation

Random Forest

- classifier_rf = RandomForestClassifier(max_depth=5, min_samples_leaf=5, n_estimators=30, n_jobs=-1, random_state=42)
- Accuracy = 98.57%
- F1 Score = 98.50

Feature importance analysis

- u_length 0.243744
- entropy 0.116133
- extension 0.096287
- dot 0.089093
- a 0.087025
- href 0.055039
- extlink 0.048834 h 0.034012
- alexa_ngram_count 0.029051

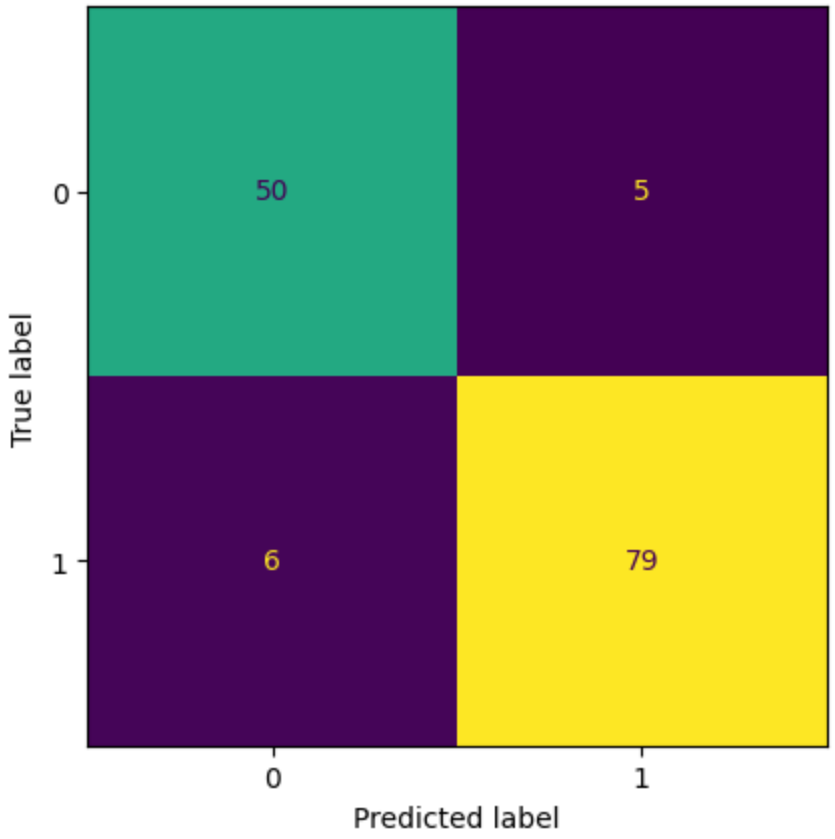


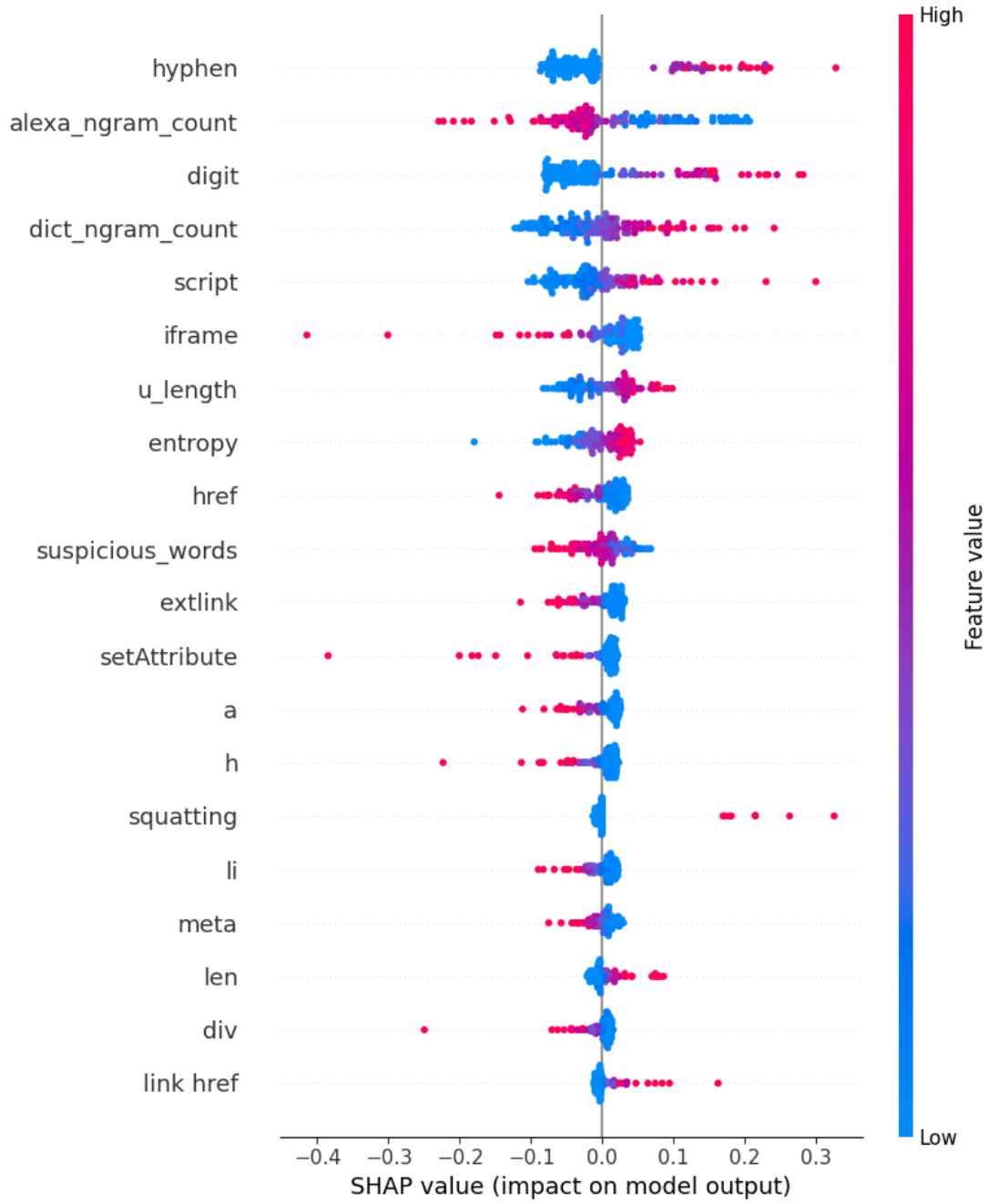
Models and Evaluation

Neural Networks

- Implemented using PyTorch
- 2 hidden layers + 1 output layer
  - ReLU & Sigmoid
  - BCE Loss
  - Dropout
  - Adam (lr = 0.001)
  - Batch Size = 32
  - Epoch = 20
- Accuracy = 92.14%
- F1 Score = 93.49

- SHAP for feature explainability




hyphen
alexa_ngram_count
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script
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div
link href
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
SHAP value (impact on model output)
High
Feature value
Low

Future Plan

- Accuracy may be misleading due to the size of the dataset
- Further dataset enchancement
- Preparing the final report